W-01: expanded current-data methods and results

What do the current analyses support in a concise local Wiki draft?

Materials and methods

Seven frozen source summaries and three attributed PNG/SVG figure pairs.

Retain source statements and task identifiers; generate the local draft and indexes inline.

Copy attributed figures unchanged and verify input/output hashes.

Scope and working assumptions

The upstream Methods scopes and nominal-condition assumptions are retained. This is an offline synthesis, without new experiments, upstream recalculation or publication approval.

Results

Integrated 7 source summaries and 3 attributed figures.

Expanded microscopy: 13 October, nominal 10 mM Ca / 5 mM Cd: C8: +2.66 ADU over 298.2s; Empty: +0.12 ADU over 299.5s; C6: +15.01 ADU over 298.4s; C1: +0.57 ADU over 299.6s; C2: +0.77 ADU over 299.6s. The C6 recording has the largest within-recording increase in this group. Candidate field-matched pre/post contrasts are C1 -10.22 ADU; C2 +1.92 ADU; Empty +0.68 ADU. Within-recording change and the field-matched treatment contrast answer different questions: the C1 field decreases across its paired acquisitions despite rising slightly within the challenged recording.

Expression evidence: The available imaging and reference analyses support a structured assessment of reporter behavior, technical conditions and reference codon composition. Low protein expression has not been identified as the cause of a weak response. Matched protein abundance, independent-culture identities and same-sample growth/viability measurements distinguish the remaining explanations.

Reference sequence: The labelled CDS spans reference bases 748-2097: 1350 nt and 450 aa; its translation matches its annotation.

Task	Evidence	Input data	Finished without more wet-lab data?
C-06	Reference coding sequence	W01_C06_summary.json	Yes: offline synthesis
D-01	Native plate-reader signals	W01_D01_summary.json	Yes: offline synthesis
F-01	Native microscopy and recording behavior	W01_F01_summary.json	Yes: offline synthesis
C-EX	Evidence review	W01_CEX_summary.json	Yes: offline synthesis
P-01	Stored PAH calculations	W01_P01_summary.json	Yes: offline synthesis
P-02	Conditional PAH models	W01_P02_summary.json	Yes: offline synthesis
V-01	Conceptual calcium reporter	W01_V01_summary.json	Yes: offline synthesis

Interpretation

The draft integrates 7 source analyses and 3 unchanged figures with numerical statements, methods and source indexes. Native imaging findings and the reference-sequence analysis now appear alongside the evidence review and the separate PAH computations.

References and reproducibility

Sharp & Li (1987), doi:10.1093/nar/15.3.1281.

Xia (2007), doi:10.1177/117693430700300028.

EMBOSS CAI manual: https://emboss.sourceforge.net/apps/cvs/emboss/apps/cai.html

Count source: https://sources.debian.org/data/main/e/emboss/6.6.0%2Bdfsg-15/emboss/data/CODONS/Eyeast_cai.cut

Original VMR ProtocolResult fields and cadmium_experiment_notes.md: R-WET-12AUG-VOYAGER / R-WET-12AUG-LAYOUT.

Chen et al. (2013), Ultrasensitive fluorescent proteins for imaging neuronal activity. https://doi.org/10.1038/nature12354

Montero Llopis et al. (2021), Best practices and tools for reporting reproducible fluorescence microscopy methods. https://doi.org/10.1038/s41592-021-01156-w

nd2 official documentation: https://talleylambert.com/nd2/

scikit-image registration documentation: https://scikit-image.org/docs/stable/api/skimage.registration.html

Nikon Perfect Focus System: near-infrared focus compensation maintains objective-to-coverslip position. https://www.microscope.healthcare.nikon.com/en_EU/resources/glossary/perfect-focus-system

Supplied 13 October lab notebook, body.b0029 (condition table), b0037-b0069 (prior qualitative observations).

Biomni methods review supplied in F01_biomni_methods_review.json; no arbitrary drift cutoff adopted.

F-01 and C-06 frozen summaries: primary methods and references are preserved in those input JSON files.

13 October notebook context: CTX02-CTX06; original source hashes and locators in F01_context_review.json.

https://github.com/Raminyazdani/igem-saarland-2026-docking-comparison/tree/3539358236841e3f95f3e9efdc6d954fe1a1eca2/tools/autodock-vina/md

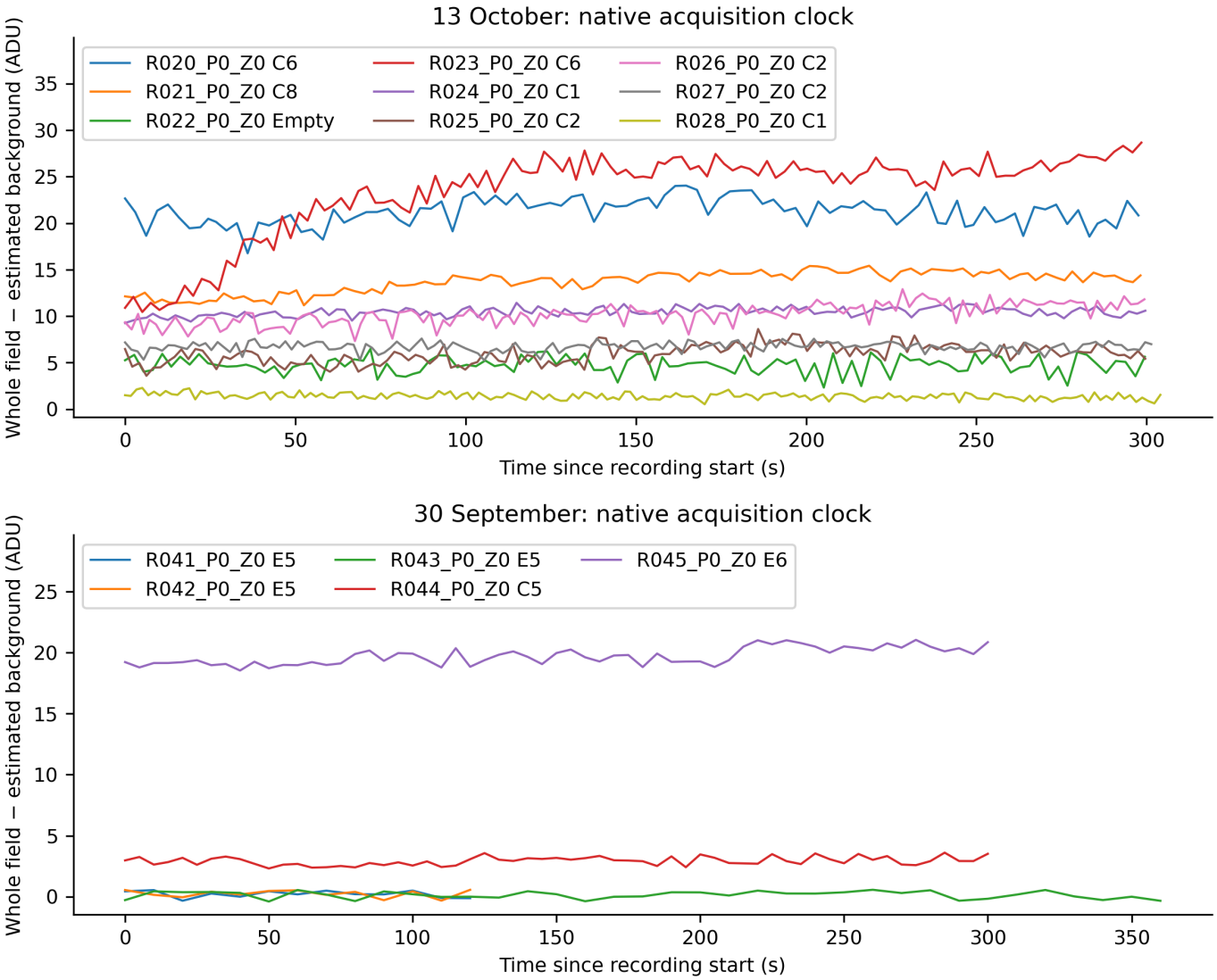
https://github.com/Raminyazdani/igem-saarland-2026-docking-comparison/tree/3539358236841e3f95f3e9efdc6d954fe1a1eca2/tools/autodock-vina/results

https://github.com/Raminyazdani/igem-saarland-2026-docking-comparison/tree/04fd02b7d71f14314549e2e0887300ea54fd5035/tools

https://github.com/Raminyazdani/igem-saarland-2026-docking-comparison/tree/3539358236841e3f95f3e9efdc6d954fe1a1eca2/tools/autodock-vina/md/models

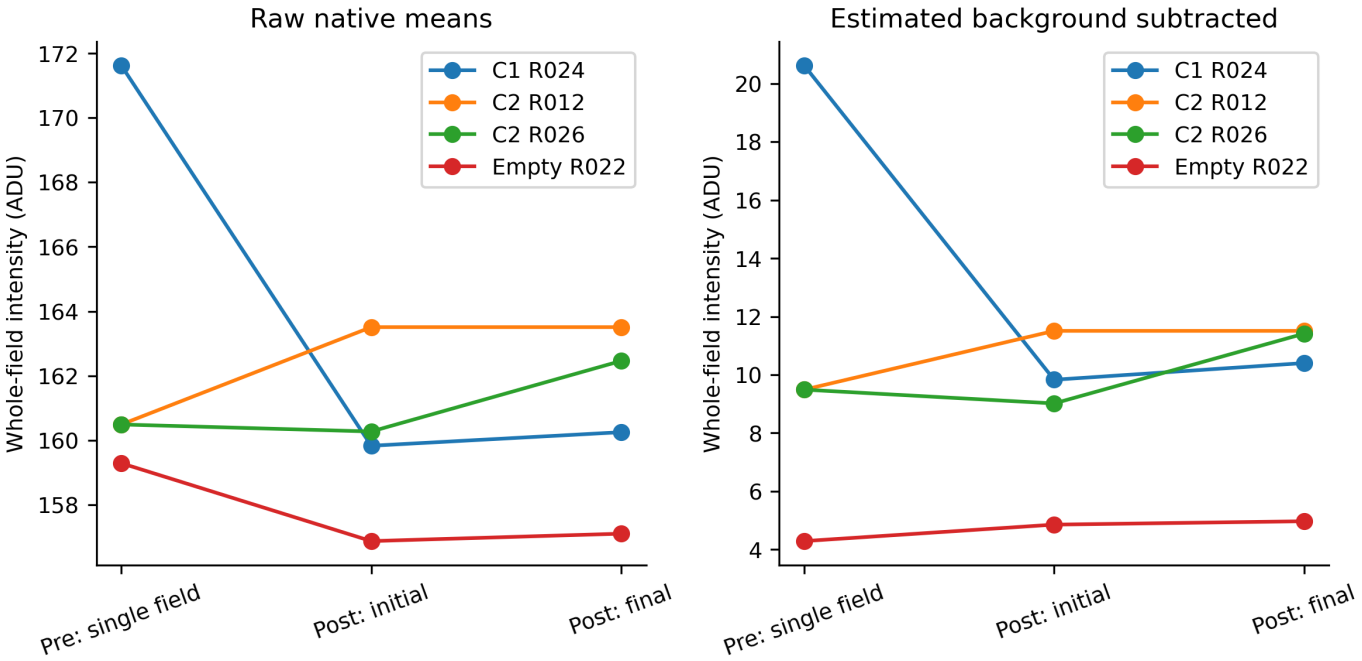
Stored source_assumptions.json: explicit source-derived parameter transcription; source hashes and complete parameters are in source_manifest.json.

Figure 1



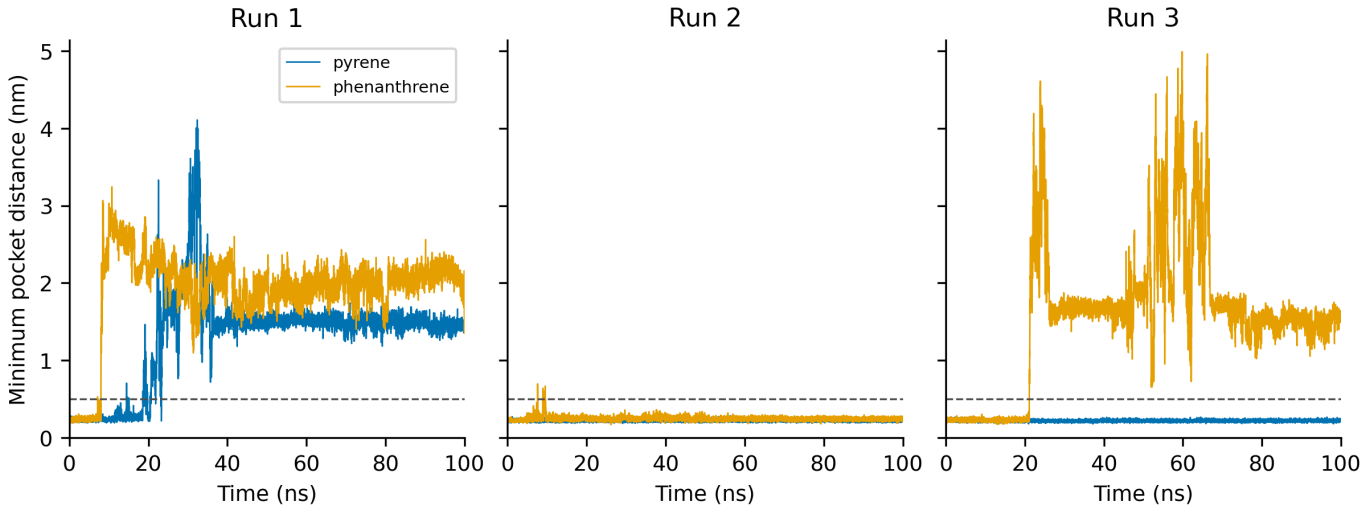
From F-01: All native time-series traces; references and positions belong to individual recordings. No Cd-addition frame is assumed.

Figure 2



From F-01: Candidate same-field comparisons pass stage-distance and registered BF-correlation checks. Post values use initial/final 10% medians; the C2 calcium-conflict static file has a single post frame. C1/Empty comparisons change both calcium and cadmium labels; C2 R026 keeps nominal calcium at 1 mM. Field matching does not establish same-cell identity, equal exposure or addition time.

Figure 3



From P-01: All six stored finite traces; dashed line is the strict 0.5 nm criterion. Run 1 initialization differs from runs 2-3.